JÖNKÖPING UNIVERSITY
School of Engineering

Techniques Used by Hackers

Lecture 6, Information Security, 7,5 ECTS

Björn Praestegaard Larsen
bjorn.larsen@ju.se

Overview

- Understand the basic methodology used by hackers
- Be familiar with some of the basic tools
- Understand the hacking mentality

What are the most common terms used by hackers?

Terminology according to ChatGPT

- What are the most common hacker terms used
- What are the most common hacker terms used within reconnaissance

Hacker

Originally, a term for someone who explores the details of programmable systems and how to stretch their capabilities. In popular media, it's often used to refer to someone who breaks into computer systems.

Hacker

- Modern hackers have become more professional and organized, turning hacking into a multi-billion dollar growth business.
- An increasing amount of hacking is state-sponsored—countries like Russia have been accused of interfering in elections, while Saudi Arabia has used state-sponsored hacking to target private businesses.
- Non-state groups of hackers often form to coordinate large-scale attacks, sometimes for activism purposes.
- Modern hacking tools include social engineering and selling malware on the dark web.
- The majority of the world's hackers live in The United States, Russia, and China.
- The goals of modern hackers typically include criminal financial gain, access to corporate or national secrets, the pursuit of political goals, or sheer fame.

Reconnaissance (Passive)

- Before an attack is executed
- Attempt to find out information
- Not directly connected to the target system.

This process is used to gather essential information without ever interacting with the target systems.

Passive scanning techniques

Detective work

- What corporate information is public?
- Who works there?
- What skills are they hiring?
- What online information can we get?
  - Netcraft site report (example of service)
    - [What's that site running? | Netcraft]

Hacker Search Engines

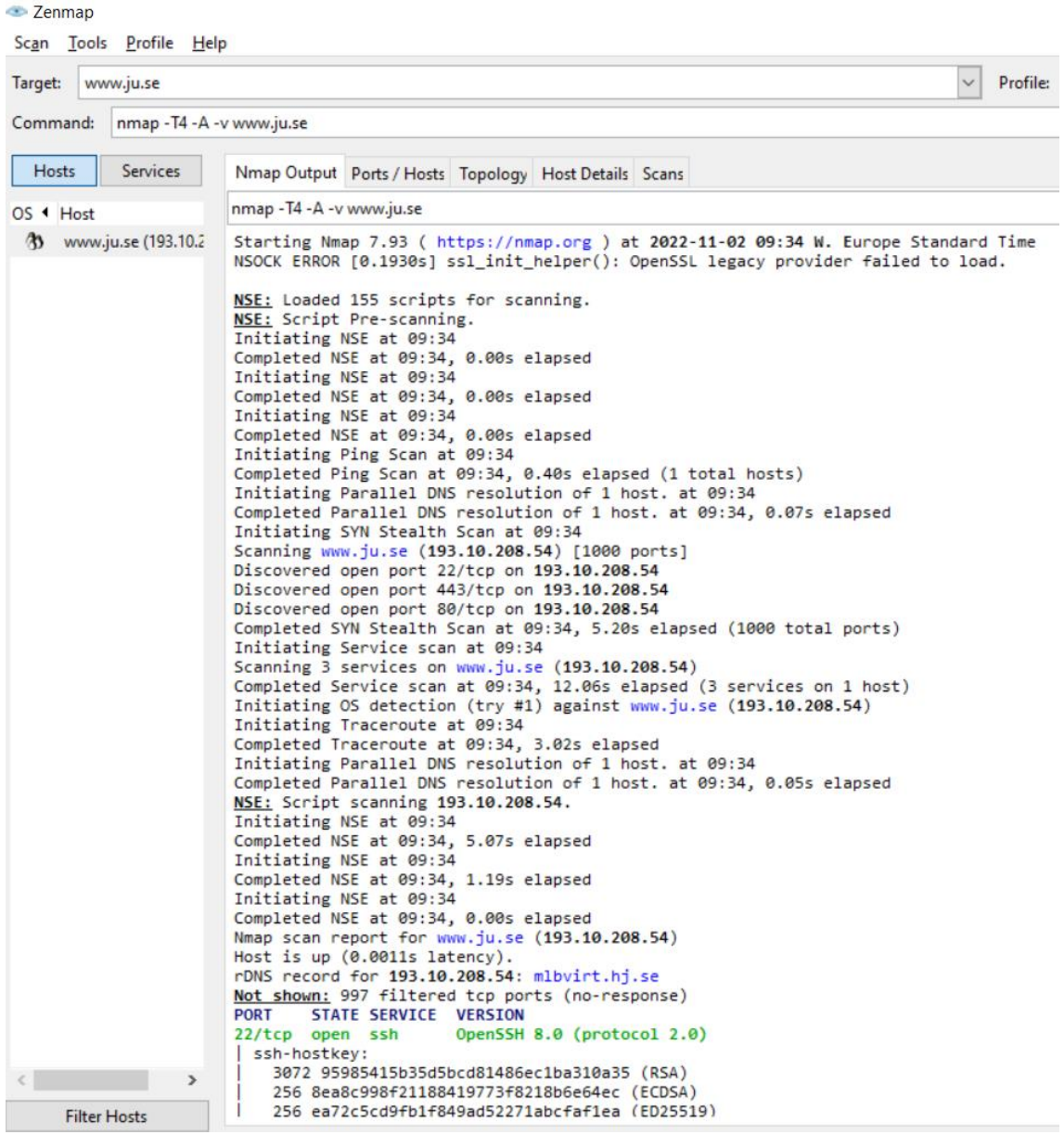
- shodan.io (Servers)
- censys.io (Servers)
- hunter.io (Email)
- urlscan.io (Websites)
- grep.app (Source code – Repository scan)
- intelx.io (Open-source Intelligence (OSINT)
- wigle.net (WiFi)
- fullhunt.io (Attack surface)
- vulners.com (Vulnerabilities)
- viz.greynoise.io (Threat Intel)
- exploit-db.com (Exploit Database CVE)
- lookup.icann.org (WHOIS Lookup)

Reconnaissance (Active)

Nmap (Port Scanning Tool)

Scans to see which ports are open

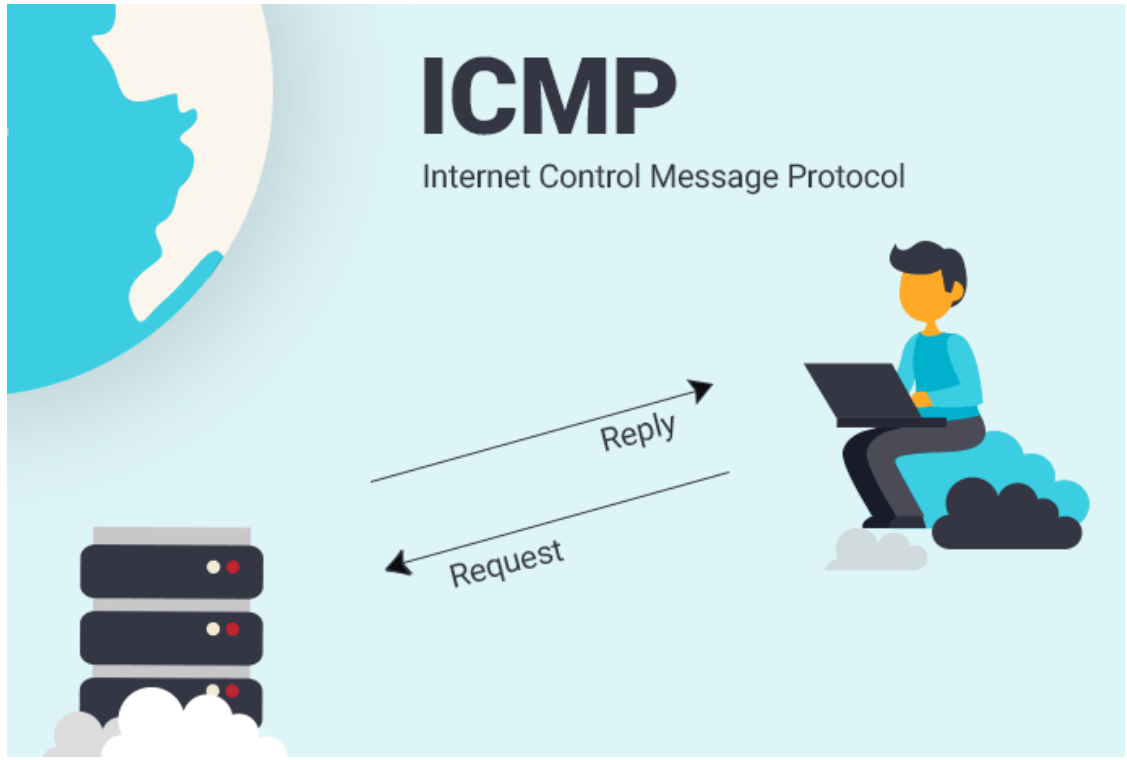
- Ping scan
- Connect scan
- SYN scan
- FIN scan



Reconnaissance (Active)
Ping scan

- Sends ICMP echo request from the source to the destination device.
- Active device returns an ICMP echo reply response.

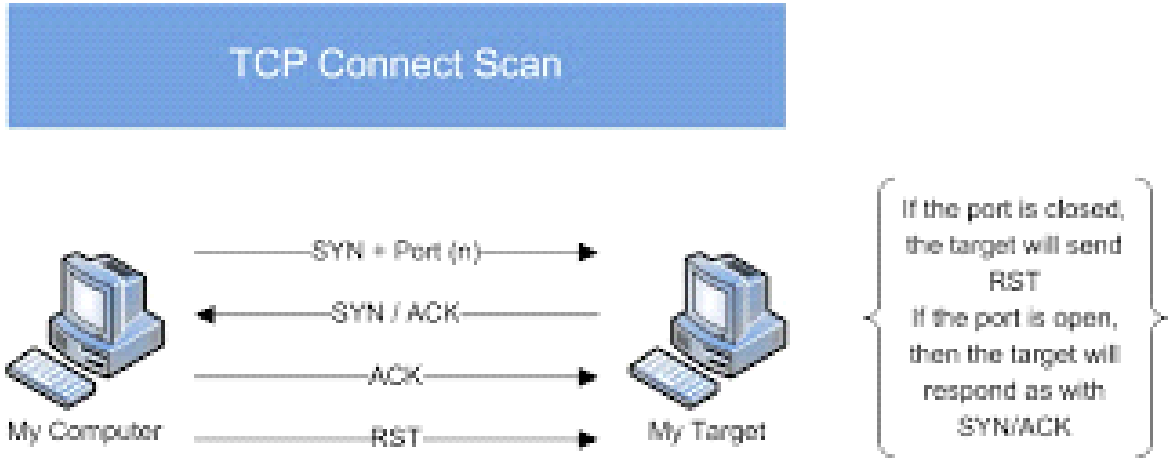
Commonly incoming ICMP packets are blocked to stop this kind of scanning.



Reconnaissance (Active)
Connect scan

- Complete connection is executed with the destination device.
- Very reliable

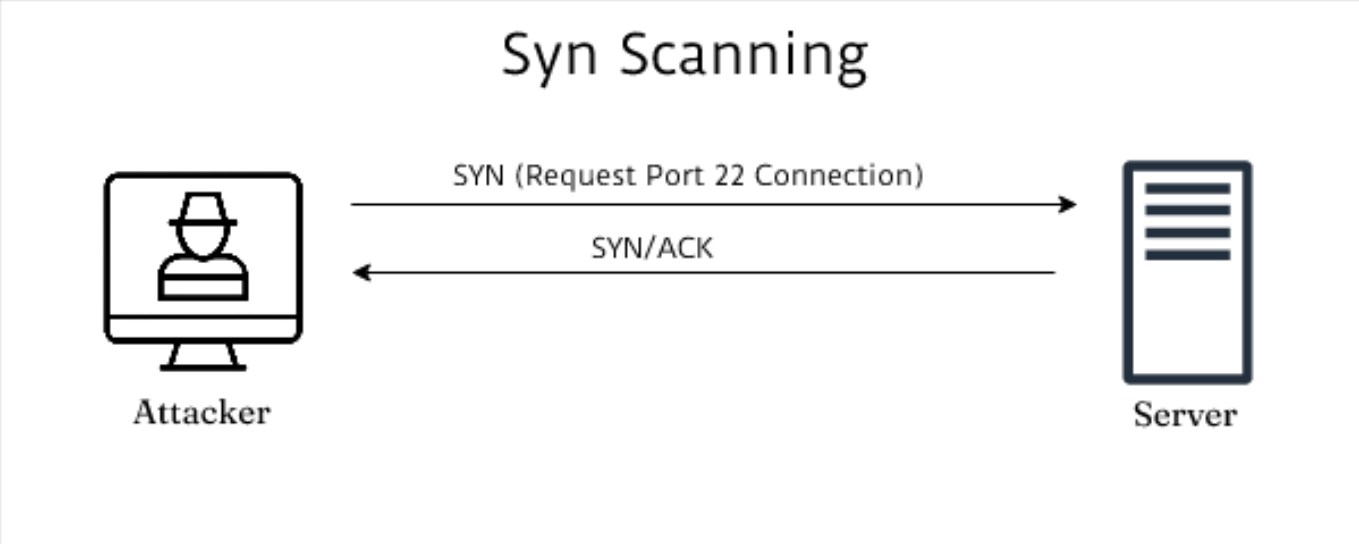
Most likely to be detected



Reconnaissance (Active)
SYN scan

- Stealthy scan
- Also called half-open scan

You send a SYN packet but you never respond to the SYN/ACK



CYBER
ATTACK
CRIMINAL
THEFT
INTERNATIONAL
INFORMATION
COMPUTER
WARFARE
CRIME
NETWORKS
INTERNET
SERVICE
UNAUTHORIZED
CYBERSPACE
FRAUD
PURPOSES
DEPARTMENT
GOVERNMENT
FACILITATED
ADVANCE
CONSIDERED
WEBSITE
GROUPS
KINETIC
RESOURCES
SURROUNDING
CITATION
TRADES
TRAFFICKERS
DIRECTLY
ELECTRONIC
HATE
CONTENT
JURISDICTIONS
BETWEEN
HACKING
AREA
ALIAS
FORM
SEVERAL
OBSCENE
DURING
HARASSMENT
PROFILE
ENTERING
PACKAGES
GOING
INCLUDE
INVESTIGATORS
COPYRIGHT
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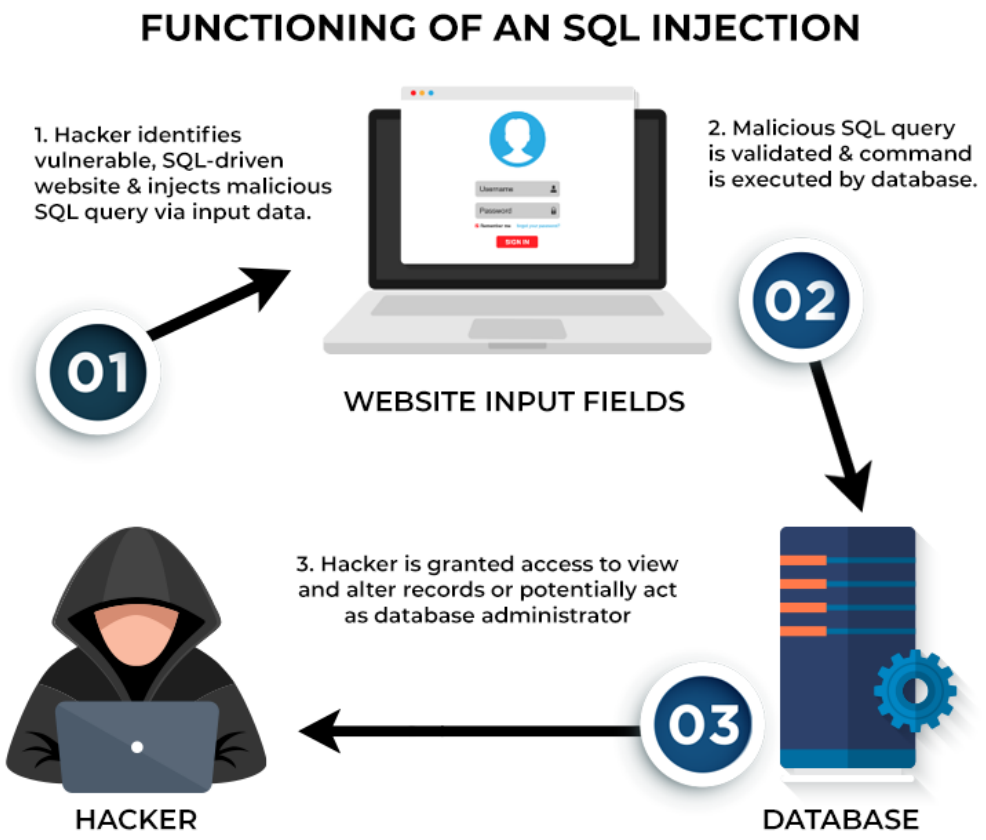
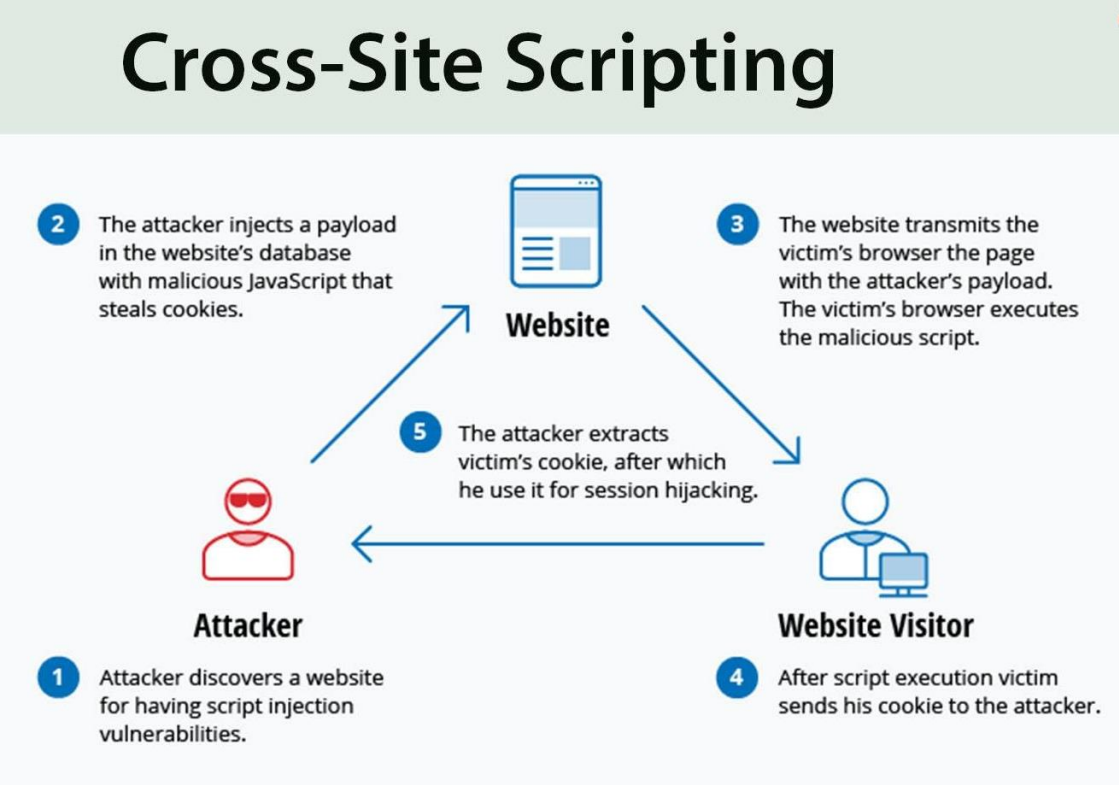
SERVICES
ABILITY
ADVANTAGE
FACE
WELL
UNSOLICITED
SYSTEM
SPAM
DEVICES
COUNTS
REFERS
ACCORDING
SPECIFIC
ACTIONS
ACTIVITIES
TARGET
MAIN
DATA
SERVER
FEDERAL
CASES
SENDING
MAIL
HARM
ORGANIZED
DOCUMENTED
BOTNET
TECHNOLOGY
COMMUNICATIONS
TERRORISM
MILLION
SCAMS
RACE
SYSTEMS
INCREASINGLY
ALTERING
CHILD
DRUG
PORNOGRAPHY
GROUP
USED
ANTI
ANALYZE
PEOPLE
CAUSE
ILLEGAL
ALLEGEDLY
UNIVERSITY
INCLUDED
NETWORK
CONCERN
HIGH
ROOMS
MAJOR
RING

SQL Script injection

- One of the most common attacks
- Depends on knowledge of SQL
- Basics are easy
- Versatile and can do a lot more than many realise
- Sends commands such as ’or’1’=‘1



Cross-Site scripting



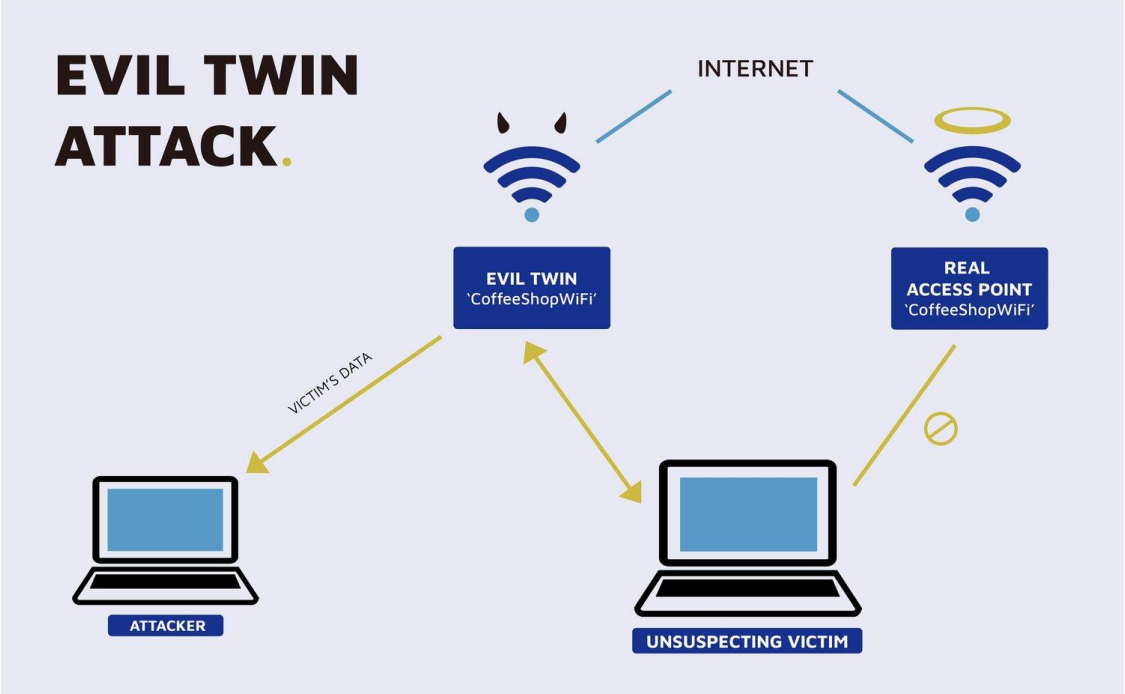
- Injects client-side scripts into webpages viewed by other users.

Education of web developers is crucial

URL Hijacking

- Purchasing close to original URL's
- Copying website
- Collecting information

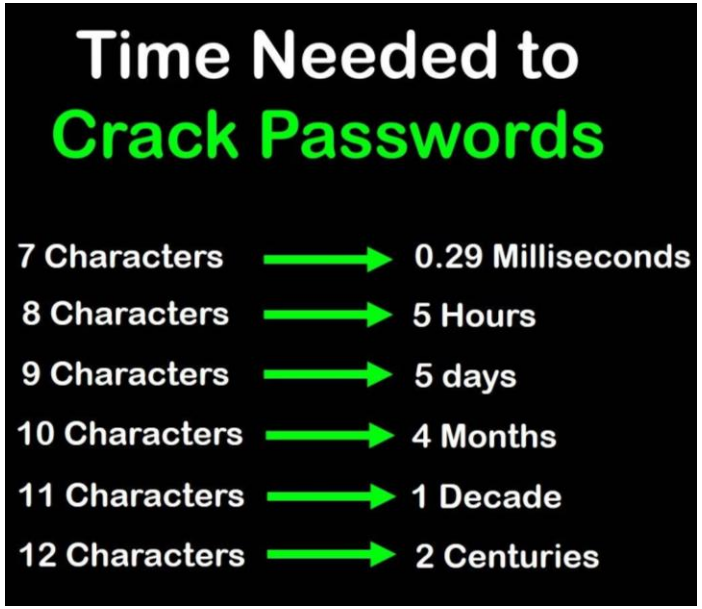
Wireless Attacks



- Evil twin attack
  - Wireless Access Point with same SSID
  - Usually busy locations with free WiFi
    - Coffee shops
    - Libraries
    - Airports

Password Cracking

- Dictionary attack
- Hybrid attack
- Rainbow table
- Brut-force



1. CrackStation
2. Password Cracker
3. Brutus Password Cracker
4. Aircrack
5. RainbowCrack
6. THC Hydra
7. Cain and Abel
8. Medusa
9. John The Ripper
10. ophCrack

According to the National Institute of Standards and Technology (NIST), password length is more important than password complexity

Rank	2021
1	123456
2	123456789
3	12345
4	qwerty
5	password
6	12345678
7	111111
8	123123
9	1234567890
10	1234567
11	qwerty123
12	000000
13	1q2w3e
14	aa12345678
15	abc123
16	password1
17	1234
18	qwertyuiop
19	123321
20	password123

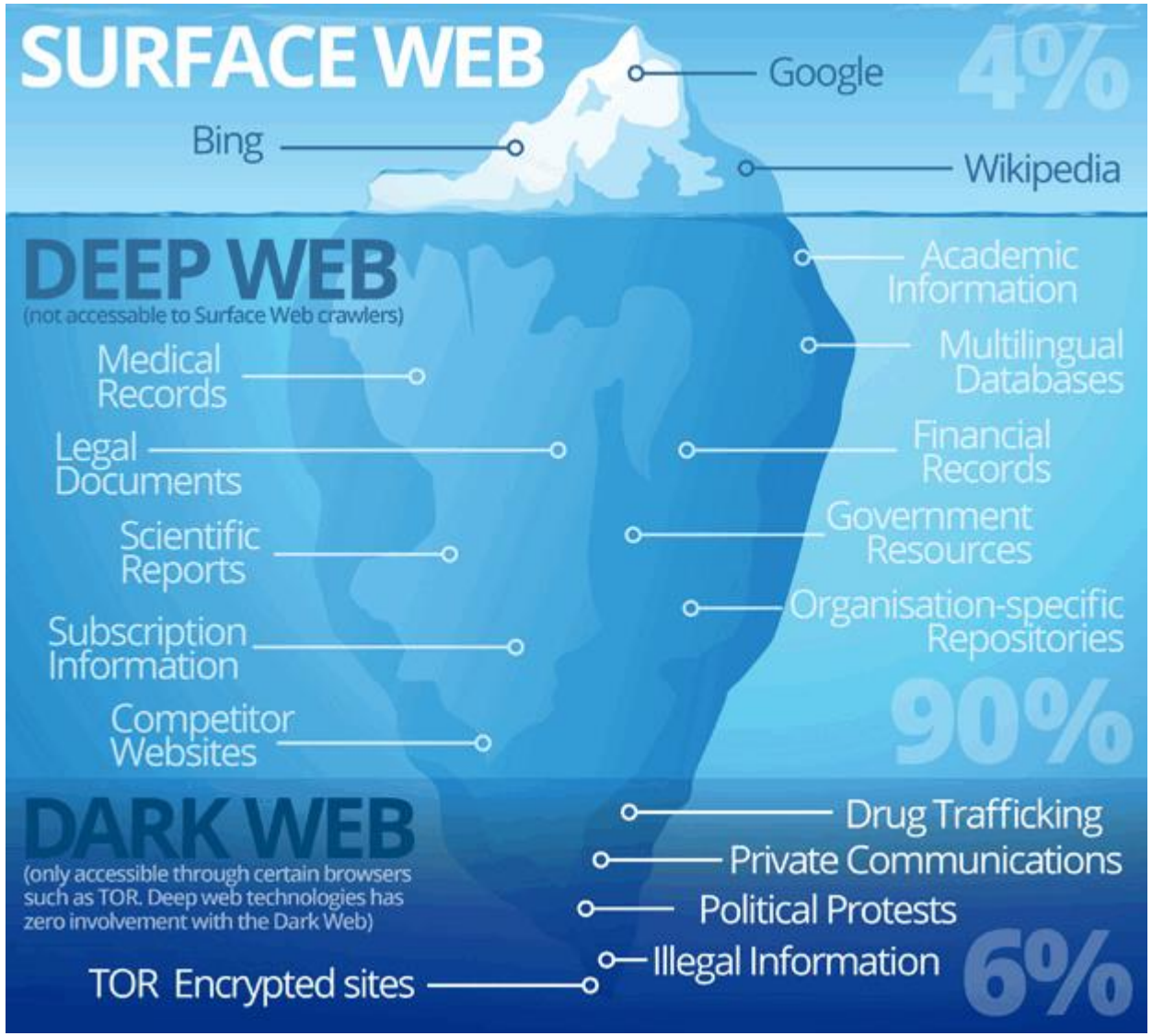
Cell phone attacks

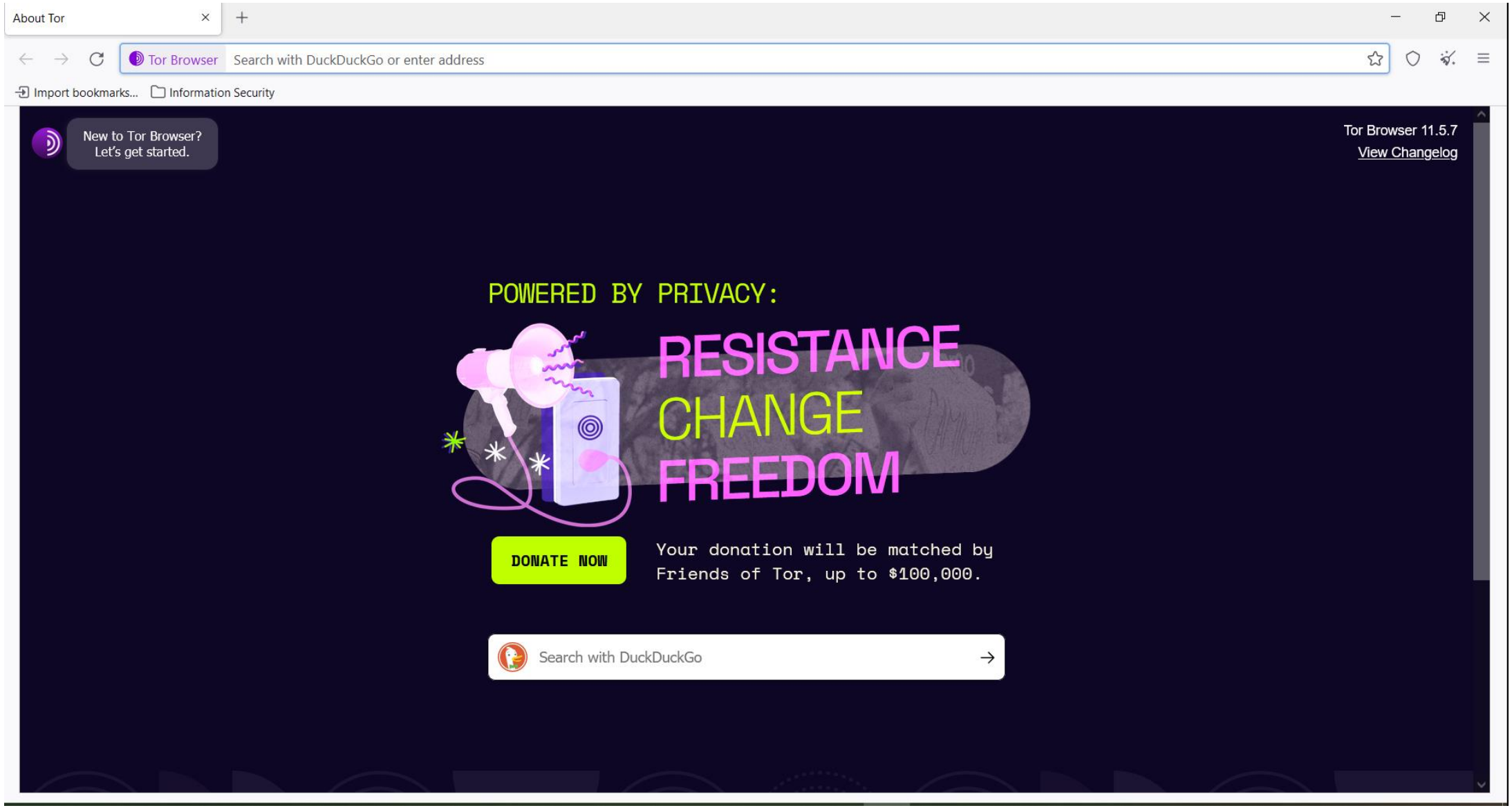
- Bluesnarfing
  - Unauthorized access of information from a Bluetooth device
- Blue jacking
  - Using another blue tooth device within range and sending messages to the target
- Bluebugging
  - Accesses and uses all phone features
- Pod slurping
  - Using a device such as an iPod to illicitly confidential data by directly plugging it into a computer where the data are held

Deep Web / Dark Web

Information is usually

- Encrypted
- Unindexed
  - Appears hidden due to lack of indexing
- Limited accessibility
  - Specific browsers needed (TOR)

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Information Security
New to Tor Browser?
Let's get started.
Tor Browser 11.5.7
View Changelog
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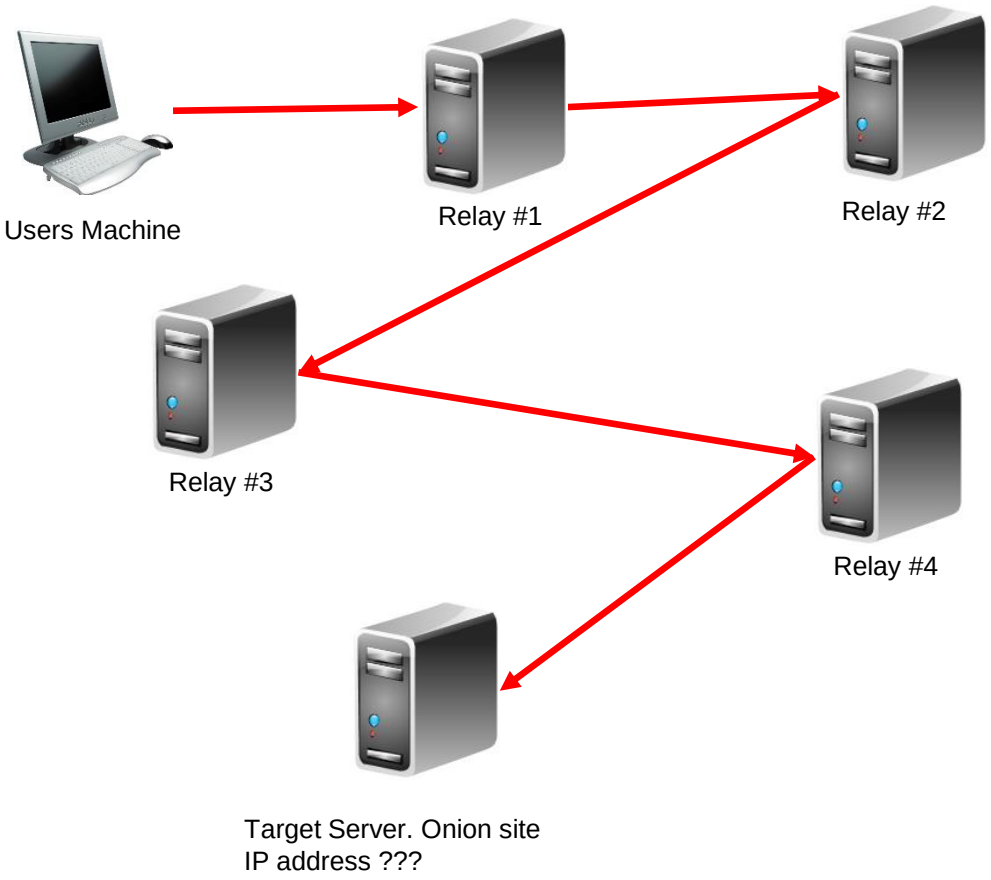
RESISTANCE
CHANGE
FREEDOM
DONATE NOW
Your donation will be matched by
Friends of Tor, up to $100,000.
Search with DuckDuckGo

Dark Web (using TOR)



- Each relay just sends the packet on and only knows the last and next hop.
- The path can change each route
- The target server only knows the last hop the packet came from
- The user only knows the first proxy in the chain

Dark Web (using TOR)

- Constructed from static index pages
- Using onion links (that looks random)
  - 2gzyxa5ihm7nsggfxnu52rck2vv4rvmdlkiu3zzui5du4xyclen53wid.onion
- Need to know what to look for

Visiting wrong links might connect you to illegal material

- Be careful -

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